



BIO 401: Quantitative Biophysics

Bio-Energetics, Enzymatic Kinetics, and Ecological Dynamics

Division of Biological Sciences & Quantitative Ecology • Fall Semester 2026

Instructor: Dr. Jordan Taylor

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Office Hours: Thursday 10:00–12:00

Lecture Hours: Mon/Wed 14:00–15:30

Room: BioSciences Hall 204

Wet Lab Session: Friday 09:00–12:00 (Lab 4)

Course Overview & Objectives

BIO 401 introduces students to mathematical and physical principles in biology. Topics include thermodynamic energy dissipation in ecosystems, Lindeman trophic transfer laws, enzyme kinetics (Michaelis-Menten, Hill cooperativity), receptor-ligand binding cascades, and non-linear population dynamics.

Grading Scheme

Assessment Component	Weight	Description
Laboratory Reports (4)	30%	Spectrophotometric enzyme kinetics and ecological field assays
Problem Sets (5)	25%	Mathematical derivations and ODE numerical simulations
Midterm Examination	20%	In-class theoretical and quantitative assessment
Final Research Project	25%	Comprehensive computational biophysics project and presentation

15-Week Curriculum Roadmap

- **Weeks 1–3:** Bio-energetics, thermodynamics of living systems, Lindeman trophic efficiency.
- **Weeks 4–6:** Trophic energy pyramids, food webs, and nutrient cycling cascades.
- **Weeks 7–8:** Single-substrate enzyme kinetics, Briggs-Haldane QSSA, and Lineweaver-Burk plots.
- **Week 9: Midterm Examination.**
- **Weeks 10–12:** Allosteric regulation, Hill cooperativity, and GPCR signal transduction cascades.
- **Weeks 13–15:** Lotka-Volterra predator-prey dynamics, phase plane analysis, and stability.